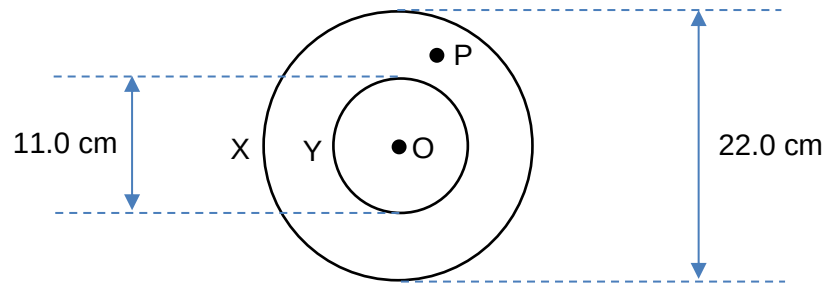


- 24** X and Y are two coaxial flat circular coils lying on a table. Coil X has 120 turns and a diameter of 22.0 cm. Coil Y has 80 turns and a diameter of 11.0 cm. O and P are two points on the table, and O is at the centre of the two coils.



Initially, there is a constant current of 1.2 A in coil X and no current in coil Y.

A current of 1.2 A is now passed through coil Y, which increases the magnitude of the magnetic flux density at P.

What is the final magnitude of the resultant magnetic flux density at O?

	A	0.14 mT	B	0.27 mT	C	0.96 mT	D	1.9 mT
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Answer: B

Since the magnitude of flux density at P increases, it implies that the currents in X and Y flow in opposite directions.

Thus the flux densities due to X and Y are in opposite directions at O.

Recall that for a flat circular coil, at its CENTRE, B is given by $\frac{\mu_0 NI}{2r}$.

$$B \text{ due to X at O} = \frac{\mu_0 NI}{2r} = \frac{(4\pi \times 10^{-7})(120)(1.2)}{2(0.220 \div 2)} = 8.22526 \times 10^{-4} \text{ T}$$

$$B \text{ due to Y at O} = \frac{\mu_0 NI}{2r} = \frac{(4\pi \times 10^{-7})(80)(1.2)}{2(0.110 \div 2)} = 1.09670 \times 10^{-3} \text{ T}$$

$$\text{Magnitude of Resultant B at O} = (1.09670 \times 10^{-3}) - (8.22526 \times 10^{-4}) = 2.7418 \times 10^{-4} \text{ T} = 0.27 \text{ mT}$$

25	A beam of beta particles enters a velocity selector. An electric field is applied in a horizontal
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